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Does Takeover Activity Cause Managerial Discipline? Evidence from International M&A Laws

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This paper exploits the staggered initiation of takeover laws across countries to examine whether the threat of takeover enhances managerial discipline. We show that following the passage of takeover laws, poorly performing firms experience more frequent takeovers; the propensity to replace poorly performing CEOs increases, especially in countries with weak investor protection; and directors of targeted firms are more likely to lose board seats following corporate-control events. Our findings suggest that the threat of takeover causes managerial discipline through the incentives that the market for corporate control provides to boards to monitor managers. (*JEL* G34, G38, K22)

Theory suggests that the threat of takeover is one of the most important external mechanisms for aligning the interests of managers and shareholders. A key prediction from the literature is that the replacement of poorly performing managers, a necessary condition for good corporate governance (Shleifer and Vishny 1989, 1997), should increase when the threat of takeover intensifies (e.g., Fama 1980; Fama and Jensen 1983; Hirshleifer and Thakor 1998).¹ This is because the threat of takeover provides additional incentives for the board to discipline managers, as board members can also be dismissed when the firm is targeted because of poor performance. Not surprisingly, a large body of empirical literature analyzes the effects of takeover activity on CEO turnover.

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¹ A survey of these models and others appears in John and Senbet (1998).

However, this literature has not been successful in establishing whether an active market for corporate control enhances managerial discipline.² Partly, this is because many studies rely on sources of variation in the threat of takeover, which can generate serious endogeneity and omitted-variable biases. For example, tests that employ the mean level of takeover activity as a proxy for the threat of takeover suffer from potential omitted-variable biases because overall takeover activity is likely to be accompanied by contemporaneous macroeconomic shocks that could jointly explain management turnover (Mikkelson and Partch 1997). Further, tests that employ takeover defenses as a proxy for takeover threats suffer from endogeneity concerns because they are often established at the discretion of the firm (Comment and Schwert 1995; Gompers et al. 2003).

To establish whether the market for corporate control improves managerial discipline, we exploit a natural experiment that increases the threat of takeover significantly: the staggered initiation of takeover acts. Takeover acts are laws passed specifically to foster takeover activity by reducing barriers to mergers and acquisitions (M&A) transactions, and thus encourage information dissemination and increase minority shareholder protection. These laws avoid the endogeneity and omitted-variable problems, to the extent that they are passed by countries and not endogenously driven by firm-specific conditions.³

Further, our international setting allows us to exploit a second source of variation to ensure that we identify the effects of M&A laws: cross-country differences in investor protection. Prior research shows that when country-level investor protection is poor, firm-level governance mechanisms are generally unavailable or prohibitively expensive (e.g., Bergman and Nicolaievsky 2007; Doidge, Karolyi, and Stulz 2007). Therefore, if the threat of takeover causes managerial discipline, the initiation of M&A laws should have a larger governance effect in countries with weak investor-protection laws, because alternative governance mechanisms at the firm level are less prevalent. Our cross-country analysis also enables us to provide unique evidence on the ability of the external market for corporate control to substitute for internal-governance mechanisms, a central question in the literature that remains unresolved (See, e.g., Pound 1992; Mikkelson and Partch 1997; Huson, Parrino, and Starks 2001). In contrast, U.S. firms are recognized to have relatively optimal internal-governance mechanisms at their disposal because of the high financial and economic development and strong investor protection in the United States (Aggarwal et al. 2009). In this way, our experiment also allows us to provide new

² Studies that find a weaker propensity to fire poorly performing CEOs during years during which takeover activity is low include Mikkelson and Partch (1997), Hadlock and Lumer (1997), and Kini, Kracaw, and Mian (2004). Studies that find that takeover activity is not related to disciplinary managerial turnover include Huson and others (2001), Denis and Kruse (2000), and Franks and Mayer (1996).

³ Staggered law changes are widely used as an instrument in U.S. and international empirical research. For example, Cornelli and others (2013) employ governance law enactments that facilitate board power in transition countries that did not have a prior legal basis for boards.

evidence on the corporate-governance effects of cross-border M&A activity and its implications for the desirability of corporate-governance reforms, as M&A laws significantly increase cross-border M&A activity (See, e.g., Bris and Cabolis 2008; Ferreira et al. 2010.). Further, by investigating how the passage of M&A laws influences the likelihood of control contests and board members' subsequent dismissal, we provide evidence on the potential mechanism that may drive the governance improvement around M&A laws.

To explore the causal effects of an increase in takeover threats and managerial discipline, we examine changes in the propensity to fire poorly performing CEOs around the time laws that increase the threat of takeover are passed. We employ a difference-in-difference methodology that measures CEO turnover, conditional on firm performance surrounding an exogenous shock to the threat of takeover. Our sample includes about 41,000 firm-year observations from 34 countries, 12 of which passed a takeover law over the 1992 to 2003 period.

Using this natural experiment, we find strong evidence that the enactment of M&A laws increases the sensitivity of CEO turnover to poor firm performance. The increase in managerial discipline caused by the initiation of M&A laws is driven by countries with weak investor-protection laws. These findings document the important role of M&A activity in ensuring good corporate-governance outcomes in countries where firm-level governance mechanisms are unavailable or prohibitively expensive. In this way, we provide evidence that the external market for corporate control, when available, can be an effective substitute for internal-governance mechanisms.

We next examine how the increased threat of takeover from the passage of M&A laws can lead to the replacement of poorly performing CEOs. At least since Fama (1980) and Fama and Jensen (1983), it has been argued that the threat of takeover should provide additional incentives for the board of directors to fire poorly performing CEOs because they themselves may be disciplined when the firm is taken over for poor performance. Hirshleifer and Thakor (1998) model this board channel and show that the threat of takeover provides a "kick in the pants" to board members to fire poorly performing CEOs to protect their own reputation and perquisites such as salary and prestige. We first show that both aggregate M&A activity increases significantly after the passage of takeover laws and that poorly performing firms are more likely to be targeted compared to other firms not subject to takeover laws. These findings confirm that the probability of a takeover attempt conditional on poor performance is greater after the passage of M&A laws. However, for the increased threat of a takeover to affect the board's diligence, there must be a penalty imposed on directors in the event of a control contest (e.g., Harford 2003). Therefore, we next examine the labor-market costs imposed on directors whose firms are targeted, and show that these directors suffer by losing their board seat in the target firm, as well as their board seats in other firms. Overall, our findings suggest that one important channel through which M&A laws increase managerial discipline in poorly

performing firms is the incentives that the market for corporate control provides to boards to monitor managers.

Our findings survive a battery of specification and robustness tests. To mitigate the concern that the passage of M&A law coincided with other changes that would independently affect the CEO-turnover-to-performance sensitivity, we control for a number of country-level developments that prior research shows can influence the propensity of firms to replace their poorly performing CEOs. These include labor laws, corporate-governance laws, stock-price informativeness, short-selling laws, insider-trading laws, financial reforms, and business cycle variation. We also examine the sensitivity of our results to the exclusion of certain observations, alternative specifications of the tests and variable definitions. Finally, it is important to note that our results are robust to controls for the change in foreign institutional investors (FIIs) surrounding the passage of takeover laws because institutional investors are associated with an active monitoring role and can facilitate cross-border M&A activity (See, e.g., Harford et al. 2011; Ferreira and Matos 2008).

Our work is also related to studies that examine manager-shareholder conflicts around the decrease in takeovers caused by the passage of antitakeover statutes in the United States. This literature shows that the decreased threat of takeover causes management to engage in more self-serving actions, including higher CEO pay, less managerial effort (the “quite life”), and lower managerial ownership and firm advantage (e.g., Bertrand and Mullainathan 1999, 2003; Cheng and Indjejikian 2009; Cheng, Nagar, and Rajan 2005). We add to this literature by focusing on the important unresolved question in the literature: does the threat of takeover cause an increase in disciplinary managerial turnover?

1. Sample Selection and Descriptive Statistics

1.1 Top-executive turnover

We gather data on executive turnover and firm performance from the *Worldscope* database between 1992 and 2003. To ensure that confounding related laws do not influence our results, we end our sample period before the implementation of the EU Takeovers Directive (2004/25/EC) in April 2004.⁴ The initial sample consists of approximately 38,000 firms from 59 countries. We exclude firms with missing firm-specific financial and executive data, firms with no identifiable top manager, and firms located in countries with missing legal environment data. We also exclude firms with assets less than \$10 million and firms that are reported in the *Worldscope* database only once, because we need at least 2 consecutive years of data on company officers and their titles to

⁴ This law allows countries considerable flexibility to opt out of the Directive and it allows for reciprocity in laws, making it more ambiguous which takeover laws are applicable across countries. In addition, the historical executive data for non-U.S. firms available through *Worldscope* CDROMs were discontinued in 2005.

compute CEO turnover.⁵ Further, we exclude countries that passed a takeover law prior to the beginning of our sample period, because there is no pre-law period for these countries. Finally, we exclude firms that were at least 50% acquired using the Security Data Corporation M&A database, because the top-manager turnover is generally higher following a corporate-control event.⁶ This screening allows us to isolate the effects of the increased threat of takeover on managerial discipline, by excluding the mechanical changes in the ownership and board structure that arise from completed M&As. The resulting sample includes 41,792 firm-year observations of 10,851 firms from 34 countries over the period 1992 to 2003 period. We have 12 countries that passed takeover laws for the first time between 1993 and 2002, which ensures we have at least one year of data for the pre-law and post-law periods. In robustness tests, we show our results hold if we restrict our sample to just these 12 countries. A breakdown of the sample distribution across countries, M&A law status, and over years is reported in Panel A of Table 1.

1.2 International M&A laws as a source of variation in the threat of takeover

We gather information on takeover acts around the world by hand collecting information from various sources, including financial law publications, such as the *International Comparative Legal Guide* and *International Financial Law Review*, studies such as Nenova (2006), and the websites of national regulatory agencies. Information on the takeover laws and their sources is reported in Table A1 in the Appendix.

Why do policymakers around the world enact legislation that promotes M&A activity? The rationales include attracting domestic and foreign capital into the stock market, increasing the country's global competitiveness, promoting more efficient capital allocation, and increasing minority shareholder protection and transparency. The legal changes associated with M&A laws are significant, because these comprehensive acts are passed to foster takeover activity by reducing barriers to M&A transactions, encouraging information dissemination, and increasing minority investor protection. They are aimed at reducing informational uncertainties regarding the legal framework applicable to M&A transactions, thus simplifying the application of various laws in connection

⁵ We follow studies such as Lel and Miller (2008), DeFond and Hung (2004), and Gibson (2003), to identify top managers within each country. Worldscope does not provide information on CEO age and tenure, and media coverage for the sample firms varies substantially across countries. Thus, as in prior studies that use similar data, we do not know whether the CEO turnover is voluntary (e.g., because of retirement) or not. However, voluntary turnovers are unlikely to be related to performance because not distinguishing between voluntary and forced turnovers causes additional noise in the dependent variable, which only affects standard errors (Hermalin and Weisbach 2003). Consistent with this argument, the evidence suggests a similar or more-sensitive relationship between CEO turnover and performance for forced replacements (See, e.g., Huson et al. 2001; Dahya et al. 2002.).

⁶ See, for example, Martin and McConnell (1991); Denis and Denis (1995); Denis, Denis, and Sarin (1997); and Denis and Kruse (2000).

Table 1
Descriptive statistics

Panel A, by country

Country	# Obs.	# Firms	CEO turnover %	Takeover law year
Argentina	36	18	25.00	None
Austria*	456	146	14.47	1998
Brazil	330	149	11.21	None
Chile*	445	148	17.75	2000
China	272	168	19.12	None
Colombia	3	3	33.33	None
Czech Republic	81	42	16.05	None
Denmark	1,047	271	13.37	None
France	2,954	993	11.68	None
Germany*	3,257	967	16.43	2002
Greece	496	211	14.52	None
Hungary	113	36	19.47	None
India*	1,119	340	12.15	1997
Indonesia*	906	298	21.85	1998
Ireland*	358	85	12.01	1997
Japan	20,440	3,733	14.04	None
Korea	1,592	653	38.13	None
Luxembourg	54	18	14.81	None
Malaysia*	1,964	663	12.37	1998
Mexico	291	114	16.84	None
New Zealand*	238	79	16.39	2001
Norway	326	103	19.02	None
Pakistan*	382	93	10.47	2000
Peru	87	43	24.14	None
Philippines*	573	186	19.02	1998
Poland	235	83	19.15	None
Portugal	146	62	8.90	None
Sri Lanka*	31	13	9.68	1995
Switzerland	1,308	349	20.03	2004
Taiwan*	1,161	399	24.55	2002
Thailand	752	253	16.09	None
Turkey	297	119	15.15	None
Venezuela	22	10	4.55	None
Zimbabwe	20	3	10.00	None
Total	41,792	10,851	15.72 (average)	-

Panel B, by takeover-law adoption

Sample	Sample description	# Obs.	CEO-turnover %	
			All firms	Top quartile – bottom quartile
(1)	Takeover law initiated during the sample period	10,890	16.31	-4.11***
(1-pre)	Before takeover law is initiated	5,767	15.47	-2.60*
(1-post)	After takeover law is initiated (treatment group)	5,123	17.26	-5.27***
(2)	Takeover law is never adopted during sample period	30,902	15.52	-3.25***
(1-pre) + (2)	The control group	36,669	15.51	-3.13***

(continued)

with M&A transactions and streamlining M&A procedures. They also prevent unnecessary and time-consuming amendments to the various other laws related to M&A activities.⁷

⁷ If there is no specific takeover act or code, it is generally the case that M&A-related issues are stipulated in various statutes, including the Companies Act, Commercial Code, Capital Markets Law, Tax Laws, and Labor Laws.

Table 1
Continued

Panel C, financial and country characteristics

	N	Mean	Median	Standard deviation
Earnings ratio	41,792	0.07	0.04	2.74
Excess stock returns (%)	37,477	−0.03	−0.07	0.49
Log assets (\$U.S. millions)	41,792	6.69	6.07	2.81
Inside-ownership %	41,792	0.35	0.36	0.28
Cross-listed	41,792	0.06	0.00	0.23
Common law	41,792	0.12	0.00	0.32
Anti-self-dealing index (ASDI)	41,792	0.48	0.48	0.16
Log GDP per capita	41,792	9.64	10.33	1.18

Table 1 presents the distribution of the sample used in the regression analysis by country, M&A law status, and year, as well as descriptive statistics for the main variables. Panel A shows the number of observations, firms, CEO-turnover rates, and the year M&A laws are passed across countries. Panel B displays the distribution of the sample and CEO-turnover frequency by M&A law status. The last column in Panel B reports the difference in CEO-turnover rates between the bottom and top quartiles of firm performance, measured by 1-year lagged earnings ratio. Panel C shows summary statistics for the sample used in the analysis. “Takeover-law year” is the year that the country passes the initial takeover law, if it has one as of the end of 2003. “Earnings ratio” is the one-year lagged ratio of earnings before interest and taxes to total assets. “Excess stock returns” is one-year lagged total investment returns in excess of the local market index. “Log assets” is the natural logarithm of the total assets of firms. “Inside-ownership %” represents the ownership stake of insiders including officers, directors and their immediate families, and shares held in trust by other companies. “Cross-listed” is an indicator variable that equals one if the firm has an American Depository Receipt issuance. “Common (civil) law” equals one for firms located in countries with an English (French, German, or Scandinavian) legal system. It is from La Porta, Lopez de Silanes, and Shleifer (2006). The “anti-self-dealing index” measures the level of difficulty for minority shareholders to thwart the consumption of private benefits by controlling shareholders, from Djankov and others (2008). “Log GDP per capita” is the natural logarithm of the GDP per capita. * refers to the country that has observations in both the pretakeover law enactment period and the postenactment period in our sample.

In terms of their comparison to U.S. state takeover laws, the M&A laws in our sample have provisions similar to the fair price laws in Connecticut, the registration and disclosure laws adopted by Hawaii, and the cash-out laws (mandatory bid thresholds) in Pennsylvania. However, they also have some unique provisions. For example, some countries introduce significant tax benefits to acquirers and new types of M&As, such as cross-border and cash-out mergers in Taiwan, whereas some allow acquirers to squeeze out minority shareholders, as in New Zealand, and some others require the potential acquirers to disclose their intent when they pass the disclosure threshold of 5% or 10%. Moreover, some simplify government approval procedures, as in Philippines, whereas others eliminate certain takeover defenses and unequal treatment of target shareholders in terms of offer price, as in India.

In summary, our experiment design allows us to mitigate the endogeneity and omitted-variable problems inherent when the level of actual takeovers is used to proxy for the threat of takeover by employing the staggered passage of international M&A laws. Our identification strategy assumes that the threats of takeover increases in the legislating countries after the laws that are designed

These regulations can be insufficient to eliminate informational asymmetries and could increase uncertainty about which law is applicable under different circumstances, especially from the point of view of foreign acquirers. Upon the enactment, the takeover act supersedes other laws and regulations concerning M&As in general.

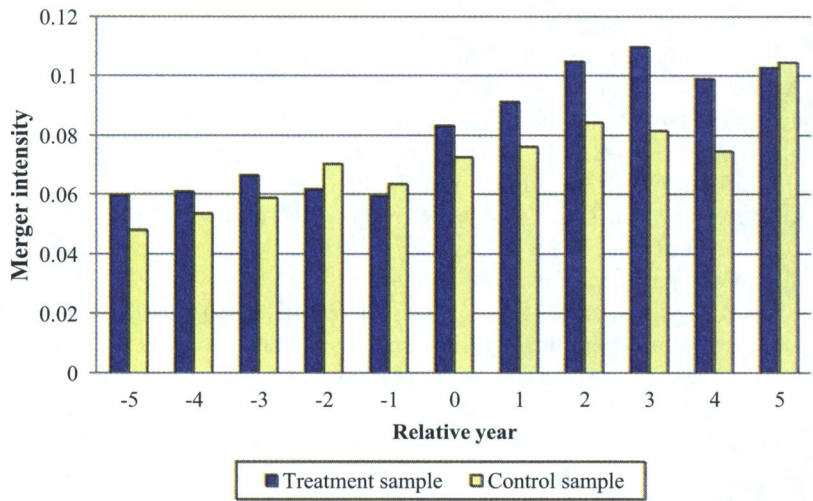


Figure 1
Takeover laws and M&A activities

Figure 1 displays the average annual merger intensity for our treatment and control sample observations. Following Rossi and Volpin (2004), we define *merger intensity* as the number of M&As divided by the total number of listed firms, calculated separately for each country and year, and then averaged across countries and relative years within the treatment and control samples. The treatment sample includes firms located in 12 countries that initiate takeover laws during the period 1992 to 2003, and the control sample includes firms from countries without takeover laws during the same period.

to promote takeovers are passed. In a series of specification tests, we show that the data support this assumption. For example, Figure 1 shows that the average merger intensity (defined following Rossi and Volpin 2004, as the number of M&As of all firms, divided by the total number of publicly listed firms in the country in a given year) increases by 2.6 percentage points following the initiation of M&A laws. Table 1A of the Online Appendix confirms this increase within a regression framework, including a set of controls. Decomposing M&A activity country by country, we find cross-border M&A activity increased for all 12 countries, and domestic M&A activity increased in 11 of 12 countries pre and post (Malaysian domestic M&A activity being the one exception). Further, we show in Section 5 that it is the acquisition of poorly performing firms that increases substantially following the adoption of M&A laws.

Further, the use of these laws solves the endogeneity and omitted-variable problems, to the extent that they are passed by countries and not endogenously determined at the firm level. To this end, it is important to note that unlike firm-level takeover defenses, countrywide M&A laws are not passed on a firm-by-firm basis. At the same time, a potential concern is that contemporaneous country-level changes could affect the turnover to performance sensitivity. We show in our later robustness section that the data do not support this alternative interpretation.

1.3 Descriptive statistics

Panel A of Table 1 provides summary statistics for the sample based on a firm's country of domicile. The average turnover ratio is 15.72%. Panel B reports turnover statistics by the presence of takeover laws. The CEO-turnover ratio increases from 15.47% in the period prior to the passage of M&A laws to 17.26% following the adoption of the laws for the 12 countries that passed takeover laws during the sample period. This increase of 1.79% is statistically significant at the 5% level (p -value = 0.017). Compared with the countries that never pass such laws, the increase in CEO turnover is 1.73%, and it is statistically significant at the 1% level. These statistics suggest that the initiation of takeover laws significantly increased the propensity of firms to replace their CEOs. The difference in CEO turnover between the 12 countries in the pre-law period and 22 countries that never pass such laws during the sample period is statistically indistinguishable from zero (0.06%), implying that our treatment group has CEO-turnover rates prior to the adoption of the M&A laws similar to those of the 22 countries.

We also compare CEO turnover between those of our treatment group of firms in the post-law period and those of the control group of firms conditional on firm performance. The last column in Panel B shows that the difference in CEO turnover for firms in the top quartile and bottom quartile of performance was similar across countries that passed M&A laws during the pre-law period (−2.60%) and countries that did not pass such laws (−3.25%). The finding thus suggests that the firms in both groups are nearly comparable with each other in terms of the sensitivity of CEO turnover to firm performance. After the passage of takeover laws, CEO turnover becomes more sensitive to poor firm performance, as the firms in the bottom quartile of firm performance are more than 100% more likely to be replaced following the enactment of M&A laws compared with firms in the top quartile in the pre-law period (−5.27% versus −2.60%). Panel C reports summary statistics of the main performance and control variables used in the regression analysis. Overall, M&A laws appear to have increased the sensitivity of CEO turnover to poor firm performance, consistent with the hypothesis that the market for corporate control strengthens disciplinary managerial turnover.

An assumption in the difference-in-difference estimation is that the treatment and control group of firms follow similar patterns prior to the event. We examine this parallel trends assumption by focusing on the sensitivity of CEO turnover to firm performance in the 3 years prior to the adoption of takeover laws. Figure 2 illustrates that we observe similar patterns in this sensitivity between countries that witnessed the passage of takeover laws and those in the control group, supporting the existence of parallel trends.⁸

⁸ Figure 2 plots the pairwise correlation coefficient estimates between CEO-turnover dummy and firm performance for our treatment and control groups of firms over the same period, and it shows a similar parallel pattern. As an

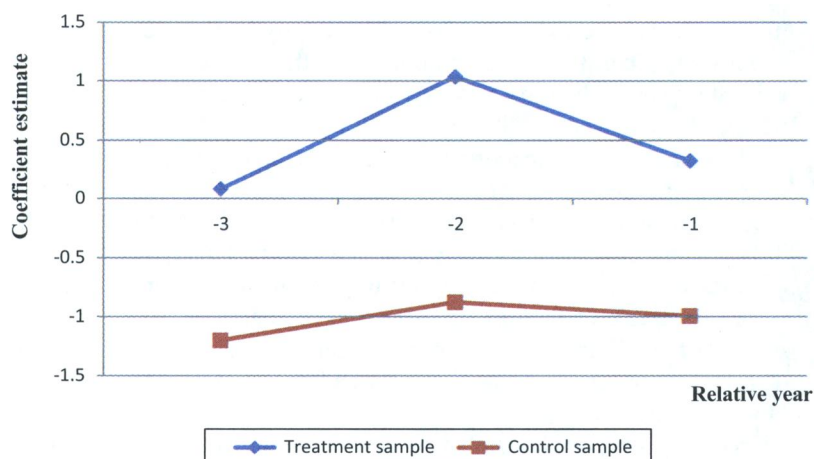


Figure 2

Parallel trends in the sensitivity of CEO turnover to firm performance

Figure 2 plots the coefficient estimates of the likelihood of CEO replacement as a function of firm performance after controlling for firm size, industry and country fixed effects, averaged across countries in the treatment and control samples. The treatment sample includes firms located in 12 countries that initiate takeover laws during the period 1992 to 2003, and the control sample includes firms from countries without takeover laws during the same period.

2. Research Design

We estimate a series of probit models to test our hypothesis that the increased takeover threat from adopting M&A laws leads to greater managerial discipline as follows:

$$\begin{aligned}
 \Pr(\text{Turnover}) = & \Phi[\alpha + \beta_1^*(\text{Firm performance}) + \beta_2^*(\text{Post}^*\text{Treat}) \\
 & + \beta_3^*(\text{Firm performance}^*\text{Post}^*\text{Treat}) \\
 & + \beta_4^*(\text{Firm performance}^*\text{Treat}) + \gamma^*X + \delta^*\text{Country FE} \\
 & + \lambda^*\text{Year FE} + \varphi^*\text{Industry FE}], \quad (1)
 \end{aligned}$$

where Φ is the standard normal cumulative distribution, *treat* refers to the group of 12 countries that pass M&A laws in the sample period; *post* refers to the post-law enactment period for these countries, and X is a set of firm and country control variables. We also include country, industry (at the two-digit Standard Industrial Classification (SIC) level), and year fixed effects in Equation (1). Our treatment group has 5,123 firm-year observations from the 12 countries

additional robustness test, we run regressions where we control for country-specific time trends and find robust results. Another assumption is exogeneity, which indicates that assignment to treatment is plausibly exogenous with respect to CEO turnover. It is unlikely that countries would enact such comprehensive laws to influence CEO turnover, and we show in Table 2A in the Online Appendix that leads and lags tests for reverse-causality effects support our assumptions.

in the post-law period and 5,767 observations in the pre-law period. The control group has 36,669 observations from the 22 countries that never adopt a takeover law in the sample period as well as the 12 countries in the pre-law period.

The complete specification of Equation (1), before redundant variables are dropped, includes *post*, *treat*, *post*firm performance*, *treat*firm performance*, *post*treat*, and *post*treat*firm performance*. The last two variables are difference-in-difference estimators of the effect of takeover laws on CEO turnover and the CEO-turnover to firm-performance sensitivity. The latter is our key variable of interest. In estimating Equation (1), the variables *post*, *treat*, *post*firm performance* become redundant and are dropped because we employ country and year fixed effects, and have a staggered adoption of the laws.⁹

We follow prior studies and measure *turnover* as a binary variable that takes on the value one if the top manager is changed in that year. We use the ratio of accounting earnings before interest and the taxes-to-book-value of assets (*earnings ratio*) as our main measure of firm performance because it is not influenced by firms' capital structure policies or by differential country-level tax regimes and the informativeness of stock prices (e.g., DeFond and Hung 2004; Huson et al. 2001; Kang and Shivdasani 1995; Volpin 2002). We lag the performance variable by 1 year, to prevent a possible overlap of the replaced CEO's performance with that of the new CEO. In later robustness tests, we also use excess stock returns as a stock market-based measure of firm performance and find similar results. We also control for the presence of large shareholders as ownership that is more concentrated may provide incentives for additional monitoring. Alternatively, ownership that is more concentrated may insulate managers with controlling stakes in their own firms (e.g., Volpin 2002). We employ the variable from *Worldscope* on the percentage of shares held by large shareholders (WS item: closely held shares). We also control for whether the firm is cross-listed in the United States, as several studies show that foreign firms cross-listed on major U.S. exchanges improve their corporate-governance standards because of being subject to stringent U.S. investor protections (e.g., Doidge et al. 2004). Previous research also shows both the percentage of closely held shares and whether a firm has an ADR are related to the level of FII holdings.¹⁰ Institutional investors are associated with an active monitoring role and can facilitate cross-border M&A activity (See, e.g., Harford et al. 2011;

⁹ *Treat* is always equal to one for countries in the treatment group, and thus it is redundant because of country fixed effects and thus is dropped. The variable *post* equals one for the years following the adoption of takeover laws. Because we have staggered adoption of the laws, this dummy variable becomes redundant to *post*treat*, as the *post* dummy takes on the value of one only for the treatment countries and in the post-law period. Because *post* and *post*treat* are the same in practice, the interaction term *post*firm performance* is redundant; thus, it, too, is dropped from the specification.

¹⁰ For example, Leuz, Lins, and Warnock (2010), among others, show that FII holdings are positively related to whether a firm has an ADR, whereas Dahlquist and others (2003) show that the percentage of closely held shares is negatively related to FII holdings.

Ferreira et al. 2008). In Table 3A of the Online Appendix, we confirm that FII holdings increase surrounding the passage of takeover laws. Moreover, in Section 4.1, we provide robustness tests with additional controls for the change in FII holdings.

Equation (1) also includes firm size measured as the natural logarithm of the book value of total assets in millions of U.S. dollars, the natural logarithm of gross domestic product (GDP) per capita, and industry dummies to control for the possible effects of firm size, industry-wide factors, and changing economic conditions on M&A activity, firm profitability, and CEO turnover. Further, we include country fixed effects, which ensure that we are controlling for unobserved country effects such as cultural attributes, for the takeover markets, as well as measuring the within-country differences in CEO turnover and firm performance.¹¹ Finally, our regressions include indicator variables for each year. Continuous variables are winsorized at the 1% level, and standard errors are corrected for possible serial correlation and heteroscedasticity by clustering at the firm level.

To analyze the effect of the market for control on managerial discipline across firms with varying degrees of investor protection, we follow a common practice in the literature and divide our sample into two subsamples based on the median values of our measures of investor protection at the country level (e.g., *Lel and Miller 2008; Leuz, Lins, and Warnock 2010*). Splitting the full sample into two subsamples based on the strength of investor protection ensures that we are fully taking into account differential effects for firm financial and governance characteristics that are shown by previous studies to exist between strong and weak investor-protection countries (e.g., *Aggarwal et al. 2011; La Porta et al. 1999*). We then test if our main variable of interest is statistically different between the two subsamples using Chow tests. This method allows us to examine our results across the country-level investor-protection measures while still controlling for important country fixed effects.

We focus on two common measures of investor protection at the country level. The first measure, from *La Porta, Lopez de Silanes, and Shleifer (2006)*, is whether the home country has an English legal origin or not, which is an overall measure of strong investor protection. The second variable is the anti-self-dealing index from *Djankov and others (2008)*, measuring the level of difficulty for minority shareholders to thwart the consumption of private benefits by controlling shareholders. *Djankov, La Porta, Lopez-de-Silanes, and Shleifer (2008)* argue that self-dealing is one of the central problems of corporate governance in most countries. We also show in Table 4A of the Online Appendix that our results are robust to other country-level measures of

¹¹ Even though we could employ firm fixed effects instead of country and industry fixed effects to exploit the panel nature of our data set fully, we do not do so because our left-hand-side variable (CEO turnover) displays relatively little time-series variation. In such cases, fixed-effects estimators lack the power to detect significant relationships, and the sample size is substantially reduced. Therefore, we follow previous studies and employ fixed-country effects (e.g., *Lel and Miller 2008; Gibson 2003*).

investor protection, such as the degree of the enforcement of laws and disclosure standards.

3. Empirical Results

3.1 Takeover laws, top-executive turnover, and investor protection

Table 2 reports the probit estimates from Equation (1). Column (1) reports results for the full sample and shows that the sensitivity of turnover to poor performance increases after the adoption of takeover laws. The coefficient on the triple-interaction term, *post*treat*firm performance*, is negative and significant (-0.851 , $t = -2.89$), consistent with the hypothesis that top managers face increased disciplinary pressure following the adoption of takeover laws. Following Ai and Norton (2003), we correct the coefficient on the interaction term for the nonlinearity of the probit specification, and continue to find a negative and statistically significant coefficient on the interaction term (-0.119 , $t = -2.73$). This result provides support for the hypothesis that increased takeover threat from enacting takeover laws leads to greater disciplinary CEO turnover.

We next split the sample by investor-protection regimes and examine the interactions between takeover-law enactment and firm performance. Firm, country, industry, and year controls are included in all regressions. Column (2) shows that in countries with civil-law tradition, where investor protection is weakest, the corrected triple-interaction term is negative and significant (-0.269 , $t = -4.03$). In terms of economic significance, the absolute probability of replacing the CEO increases by 3.36% in the presence of takeover laws when we move from the top quartile to the bottom quartile of firm performance and compared to the control group. For comparison, Huson and others (2001) show that going from the top quartile to the lowest quartile in EBIT/TA ratio increases the probability of CEO turnover by 2%. In contrast, Column (3) shows that for common-law countries, where investor protection is strongest, the triple-interaction term is statistically insignificant. Therefore, in poor-investor-protection countries, the adoption of takeover laws improves the CEO-turnover to poor-firm-performance sensitivity. Further, the difference in the interaction terms between civil- and common-law countries is statistically significant ($p\text{-value} < 0.01$).

Columns (4) and (5) report subsample results for splits based on the anti-self-dealing index. We use the sample median to classify firms into strong and weak investor protection.¹² The results show that the corrected triple-interaction term is negative and significant in weak-investor-protection countries (-0.182 , $t = -2.32$) and insignificant in other countries (-0.054 , $t = -1.05$). Alternative

¹² If we use the average value of the anti-self-dealing index across countries used by Djankov and others (2008), our results remain robust. The only difference in cut-off points between our sample and that of Djankov and others (2008) is South Korea, which has 1,592 firm-year observations and makes up 3.8% of our regression sample.

Table 2
CEO-turnover and -takeover acts

Variables	Full sample (1)	Civil law (2)	Common law (3)	Low ASDI (4)	High ASDI (5)
Post*treat	0.258*** [3.495]	0.384*** [4.289]	0.073 [0.392]	0.401*** [3.356]	0.149 [1.579]
Post*treat*firm performance	-0.851*** [-2.896]	-1.775*** [-3.936]	0.640 [1.404]	-1.190** [-2.361]	-0.556 [-1.510]
Treat*firm performance	0.548 [1.435]	0.917* [1.905]	0.844 [0.929]	0.347 [0.860]	2.569*** [5.905]
Firm performance	-0.294 [-0.907]	-0.258 [-0.836]	-1.033 [-1.178]	-0.05 [-0.344]	-2.382*** [-6.581]
Log assets	0.066*** [10.503]	0.064*** [9.694]	0.085*** [4.059]	0.062*** [6.420]	0.079*** [9.132]
Inside-ownership %	0.482*** [8.017]	0.482*** [7.531]	0.568*** [2.948]	0.293*** [3.438]	0.739*** [8.342]
Cross-listed	0.069 [1.204]	0.089 [1.459]	0.019 [0.094]	-0.041 [-0.375]	0.131* [1.918]
Log GDP per capita	0.244 [0.571]	0.099 [0.223]	0.74 [0.624]	0.447 [0.672]	0.14 [0.227]
Constant	-3.908 [-1.019]	-2.459 [-0.616]	-5.098 [-0.712]	-5.318 [-0.890]	-4.019 [-0.743]
Country, industry, year FE	Yes	Yes	Yes	Yes	Yes
Observations	41,792	36,928	4,780	13,967	27,800
Pseudo R-squared	0.038	0.038	0.052	0.072	0.028
Mean interaction effect	-0.119*** [-2.731]	-0.269*** [-4.039]	0.075 [1.360]	-0.182** [-2.327]	-0.054 [-1.058]

Table 2 presents probit estimates of the effect of takeover laws on the relationship between the probability of CEO turnover and firm performance under various legal environments. The sample includes countries that enact a takeover law for the first time and countries that never pass such laws during the sample period. “Treat” refers to the group of 12 countries that pass M&A laws in the sample period and “post” refers to the post-law-enactment period for these countries. “Firm performance” is the one-year-lagged ratio of earnings before interest and taxes to total assets. Country, industry (at the two-digit SIC level), and year fixed effects are included in every column. The continuous variables are winsorized at the 1%-level, and time-varying independent variables are lagged 1 year. The “interaction effect” is defined as the change in the predicted probability of CEO turnover for a change in both the firm-performance and the post*treat dummy variables using the methodology of Ai and Norton (2003). Other variables are defined in Table 1. The z-statistics appear in brackets, below parameter estimates. Robust standard errors are estimated using Rogers’s method of clustering by firm. Asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

measures of investor-protection laws from La Porta and others (2006), such as the burden of proof and disclosure standards, produce similar conclusions. These are reported in Table 4A of the Online Appendix. These results again suggest that the threat of takeover improves the corporate-governance environment of firms in weak-investor-protection countries.

Overall, Table 2 shows that the enactment of takeover laws leads to an increased probability of terminating poorly performing CEOs. These results are consistent with the findings of Mikkelson and Partch (1997), Hadlock and Lumer (1997), and Martin and McConnell (1991), whose findings suggest that in the United States the threat of takeover works as a disciplinary device. Our results also show that the effects of the threat of takeover are primarily concentrated in weakly governed firms. This is consistent with the findings of

Kini and others (2004) for U.S. firms, as firms in countries with weak investor protection tend to have inadequate internal-governance practices, such as a lack of independent boards (See, e.g., Doidge et al. 2007.).

3.2 Confounding channel tests

To mitigate the concern that M&A law adoption coincided with other country-level changes that would independently affect the CEO-turnover-to-performance sensitivity, we investigate several proxies for potential confounding channels. First, we add three controls for country-level developments that prior research shows can influence the propensity of firms to replace poorly performing CEOs. These include differences in labor laws (See, e.g., Atanassov and Kim 2009.), corporate-governance laws (See, e.g., Dahya et al. 2002.) and stock-price informativeness (See, e.g., DeFond and Hung 2004.). “Labor Union” is a measure of the legal protection of labor unions and the regulation of collective disputes, obtained from Atanassov and Kim (2009). Higher values represent stronger union power. Corporate-governance laws are taken from the list of countries that passed a corporate-governance reform either on a mandatory or comply-or-explain basis, obtained from Kim and Lu (2013). We create an indicator variable to represent the adoption of such codes from the year of their passage on. The stock returns synchronicity measure, from Morck, Yeung, and Yu (2000), proxies for the degree of information available in financial markets about the firm. Higher values of this variable represent less-informative stock prices.

We also include controls for short-selling laws and insider-trading laws. Prior research shows that these laws can affect the governance environment of firms and therefore may influence CEO turnover indirectly through this governance channel. Regulations on short selling are obtained from Jain and others (2013). Following their study, we create a variable called *restriction*, which equals zero when there are no restrictions on short selling in the home country in a given year, and three when there is a total ban. From Bhattacharya and Daouk (2002), we include a dummy variable that equals one if insider-trading laws are enforced in the country of origin, and zero otherwise. We employ the financial-reform index from Abiad and others (2010), which is a broad measure of the multidimensional nature of financial reforms.¹³ We use the annual changes in this index as our measure of changes in financial reforms, where higher values represent more financial liberalization. Finally, we control for GDP growth, defined as the annual percentage change in GDP, obtained from the World Bank.

Table 3 reports results from tests where we include these variables and their interaction term with firm performance except for GDP growth, which is

¹³ It combines financial policy changes in seven categories: credit controls and reserve requirements, interest rate controls, entry barriers, state ownership, policies on securities markets, banking regulations, and restrictions on the capital account. It is normalized between zero and one and it varies over time.

Table 3
CEO turnover, M&A laws, and changes in economic and regulatory environment

Variables	Full sample	Civil law	Common law	Low ASDI	High ASDI
	(1)	(2)	(3)	(4)	(5)
Post*treat	0.232*** [2.722]	0.287** [2.509]	0.059 [0.303]	0.204 [1.245]	0.062 [0.582]
Post*treat *firm performance	-0.878** [-2.394]	-2.056*** [-3.244]	0.678 [1.357]	-1.598** [-2.003]	-0.303 [-0.716]
Treat *firm performance	1.043** [2.558]	1.511*** [3.126]	1.708 [1.418]	1.208** [2.485]	2.307** [2.328]
Firm performance	-2.903 [-0.572]	-3.396 [-0.690]	-11.640 [-0.557]	-6.972 [-1.415]	7.104 [0.919]
Log assets	0.073*** [11.132]	0.072*** [10.297]	0.086*** [4.166]	0.077*** [7.100]	0.078*** [9.050]
Inside-ownership %	0.580*** [8.935]	0.586*** [8.377]	0.587*** [3.039]	0.448*** [4.537]	0.755*** [8.419]
Cross-listed	0.047 [0.777]	0.068 [1.049]	0.004 [0.024]	-0.193 [-1.421]	0.148** [2.144]
GDP growth	-0.011 [-1.452]	-0.016 [-1.532]	-0.004 [-0.303]	0.002 [0.099]	-0.023** [-2.368]
Financial reforms	1.495** [2.048]	2.051** [2.227]	0.458 [0.307]	2.134 [1.474]	2.424** [2.559]
Financial reforms*firm performance	-3.748 [-0.840]	-8.738 [-1.386]	-1.724 [-0.186]	-0.575 [-0.065]	-9.025 [-1.632]
Short-selling restrictions	0.039 [0.462]	0.148 [1.294]	-0.018 [-0.073]	0.116 [0.527]	0.168 [1.628]
Short-selling restrictions*firm performance	-0.182 [-0.840]	-0.345 [-1.580]	1.662 [1.619]	0.083 [0.306]	0.291 [0.996]
Corporate-governance codes	-0.114 [-1.182]	0.047 [0.347]	0.046 [0.174]	0.479*** [2.589]	-0.546*** [-4.194]
Corporate-governance codes*firm performance	0.163 [0.366]	0.444 [0.665]	0.122 [0.159]	0.676 [0.836]	0.424 [0.632]
Insider-trading laws	0.536*** [2.618]	0.507** [2.459]	-2.393 [-1.100]	0.500** [2.219]	-
Insider-trading laws*firm performance	1.206 [0.538]	1.501 [0.664]	19.943* [1.816]	2.147 [0.938]	-
Labor union regulations*firm performance	-1.319 [-0.795]	-1.670 [-0.683]	8.141 [1.135]	0.568 [0.220]	-2.508 [-0.864]
Stock-price informativeness*firm performance	0.024 [0.373]	0.034 [0.633]	-0.223 [-0.761]	0.057 [0.932]	-0.127 [-1.194]
Log GDP per capita	0.158 [0.263]	-0.611 [-0.916]	1.289 [1.241]	0.576 [0.741]	0.027 [0.028]

(continued)

included as a stand-alone variable in the regression specification.¹⁴ Consistent with our previous findings, we continue to find in these tests that takeover laws increase the turnover of poorly performing top managers in countries with weak investor protection. Among the new control variables, we also find that financial reforms and the enforcement of insider-trading laws increase CEO turnover. Overall, these series of tests suggest that when we control for a number of additional country-level changes that prior theoretical and empirical literature

¹⁴ We include these variables and their interactions simultaneously but note that the results are similar if they are included in the models one at a time. “Labor Union” and “Stock Returns Synchronicity” are time invariant, and thus the stand-alone forms of these variables are dropped from the specification that includes country fixed effects.

Table 3
Continued

	Full sample	Civil law	Common law	Low ASDI	High ASDI
Variables	(1)	(2)	(3)	(4)	(5)
Constant	−4.672 [−0.757]	3.255 [0.477]	−6.275 [−0.946]	−8.425 [−1.051]	−3.402 [−0.399]
Country, industry, and year FE	Yes	Yes	Yes	Yes	Yes
Observations	38,028	33,215	4,729	10,461	27,528
Pseudo R-squared	0.026	0.026	0.055	0.042	0.029
Mean interaction effect	−0.139** [−2.232]	−0.316*** [−3.371]	0.098 [1.165]	−0.216** [−2.227]	−0.032 [−0.563]

Table 3 presents probit estimates of the effect of takeover laws on the relationship between the probability of CEO turnover and firm performance under various legal environments, while controlling for the effect of aggregate economic activity and regulatory changes. “GDP growth” is the annual percentage change in GDP, obtained from World Bank. “Financial reforms” represent annual changes in the financial-reform index from Abiad and others (2010). It is a broad measure of financial policy changes in credit controls and reserve requirements, interest rate controls, entry barriers, state ownership, policies on securities markets, banking regulations, and restrictions on the capital account. Moreover, it is normalized between zero and one. Short-selling restrictions equals zero when there are no restrictions on short selling in the country and three when there is a total ban; it is obtained from Jain and others (2013) and it incorporates prohibitions on naked short selling, up-tick rules, and total bans on short selling. “Corporate-governance codes” is a dummy variable that equals one for years following the enactment of mandatory or comply-or-explain corporate-governance codes, if any, and zero otherwise. It is from Kim and Lu (2013). Insider-trading laws equals one if insider-trading laws are enforced in the country of origin and zero otherwise, and is obtained from Bhattacharya and Daouk (2002). “Labor union regulations” measure the legal protection of labor unions and the regulation of collective disputes, obtained from Atanasov and Kim (2009). Finally, “stock-price informativeness” represents the stock return synchronicity as a proxy for the degree of information available in financial markets about the firm; these are obtained from Morck, Yeung, and Yu (2000). Other variables are defined in Table 1 and Table 2. Country, industry (at the 2-digit SIC level) and year fixed effects are included in every column. The z-statistics appear in brackets below parameter estimates. Robust standard errors are estimated using Rogers’s method of clustering by firm. Asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

suggest could also influence CEO-turnover sensitivity during our sample, our results continue to hold. These findings, taken together with the inclusion of other controls in our models, provide evidence that that our results are not being driven by other country-level changes that would independently affect CEO-turnover-to-performance sensitivity.

4. Robustness Tests

4.1 Foreign institutional-investor holdings

Our results so far show that the initiation of M&A laws leads to an increase in the threat of takeover and a corresponding increase in disciplinary managerial turnover. A potential alternative explanation is that rather than the increased threat of takeover causing an increase in disciplinary managerial turnover, the change in foreign institutional ownership around the passage of M&A laws drives our findings. Prior research suggests that institutional investors are associated with an active monitoring role and can facilitate cross-border M&A activity (See, e.g., Harford et al. 2011; Ferreira et al. 2008).

Even though all of our tests so far include controls for the percentage of closely held shares and whether the firm has an ADR that has been shown to be related to the level of FII holdings, we next compute a direct proxy for the change in FIIs' holdings. We gather data from the Thomson Reuters Global Share Ownership database. The database provides information on global equity-ownership information, covering institutions, mutual funds, and beneficial stakeholders for more than 50,000 global companies. The institutional-ownership data coverage starts in 1997, but some of the data are backfilled to 1995. We are able to match 65% of our sample with the holdings database. The results from these tests are reported in the first two columns of Panel A of Table 4.¹⁵ They show our results, which are robust after adding controls for the percentage of closely held shares, whether the firm has an ADR, and the change in FII holdings.¹⁶ They also show that FIIs are associated with improvements in the governance environment of firms in weak-investor-protection countries, as indicated by the interaction term (-0.052 , $t = -1.679$), which is consistent with Aggarwal and others (2011), who find that FIIs are more important in prompting governance changes in such countries.

4.2 Stock market-based measure of firm performance

We use the 1-year lagged earnings ratio as our firm-performance measure in main tests because stock market-based measures are less informative in some countries, owing to high stock-price synchronicity and illiquidity (e.g., Morck, Yeung, and Yu 2000). With this caveat in mind, we examine the sensitivity of our results to stock returns measured by 1-year lagged total investment returns in excess of the local market index as a further robustness test. Results reported in Columns (3) and (4) show that the adoption of takeover laws improves the sensitivity of CEO turnover to excess stock returns, and this effect is driven by firms from civil-law-based countries.

4.3 Relative industry-performance measure

Morck, Shleifer, and Vishny (1988) and Parrino (1997) suggest that internal monitors use relative industry-performance measures to evaluate CEO performance. To examine the robustness of our results to this alternative definition of *firm performance*, we recompute our firm-performance measures in which industry-adjusted performance is calculated as firm performance minus the median value of the corresponding two-digit SIC global industry. The Columns (5) and (6), in Panel A, show that the corrected interaction effect between takeover law and industry-adjusted firm performance is negative and statistically significant in countries with weak investor protection.

¹⁵ In this table, we focus on the broad investor-protection measure based on the English legal origin classification, but the classification based on the anti-self-dealing index produces similar results.

¹⁶ We exclude observations with missing FII holdings data. We obtain robust results when we instead replace missing observations with a zero and an indicator variable for missing FII data. Further, we also obtain robust results when we perform these tests using just the 12-country subsample (see Table 5A in the Online Appendix).

Table 4
Robustness tests

Panel A, controlling for changes in foreign institutional ownership and alternative measures of firm performance

Variables	Controlling for changes in foreign-institutional-ownership %		Excess stock returns		Relative industry performance	
	Civil law	Common law	Civil law	Common law	Civil law	Common law
	(1)	(2)	(3)	(4)	(5)	(6)
Post*treat	0.361*** [3.181]	0.593** [2.170]	0.328*** [3.612]	0.108 [0.553]	0.295*** [3.411]	0.105 [0.572]
Post*treat*firm performance	-1.845*** [-3.003]	0.801 [1.028]	-0.280** [-2.058]	-0.276 [-1.219]	-1.744*** [-3.865]	0.700 [1.498]
Treat*firm performance	0.951 [1.589]	2.987** [2.401]	-0.032 [-0.280]	0.908*** [3.262]	0.892* [1.855]	0.657 [0.690]
Firm performance	-0.072 [-0.348]	-2.891** [-2.669]	-0.027 [-0.713]	-0.677*** [-3.392]	-0.246 [-0.815]	-0.89 [-0.965]
Log assets	0.087*** [6.292]	0.140*** [2.820]	0.064*** [8.900]	0.110*** [4.649]	0.064*** [9.677]	0.085*** [4.059]
Inside-ownership %	0.522*** [6.035]	0.490** [1.984]	0.445*** [6.317]	0.562*** [2.735]	0.480*** [7.506]	0.567*** [2.946]
Cross-listed	0.056 [0.643]	-0.014 [-0.055]	0.088 [1.422]	-0.115 [-0.632]	0.089 [1.460]	0.017 [0.087]
Log GDP per capita	-1.607* [-1.691]	1.979 [1.139]	0.087 [0.179]	0.468 [0.299]	0.099 [0.223]	0.742 [0.625]
Change in foreign institutional-ownership %	0.011* [1.669]	-0.041 [-1.409]	-	-	-	-
Change in foreign institutional-ownership % *firm performance	-0.052* [-1.679]	0.238 [1.192]	-	-	-	-
Constant	13.169 [1.532]	-16.150 [-1.431]	-2.289 [-0.524]	-4.255 [-0.451]	-2.471 [-0.619]	-5.055 [-0.705]
Country, industry, and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	23,265	2,676	33,149	4,209	36,928	4,780
Pseudo R-squared	0.031	0.070	0.038	0.056	0.038	0.052
Mean interaction effect	-0.269*** [-3.204]	0.102 [1.334]	-0.047** [-2.473]	-0.023 [-1.264]	-0.257*** [-4.029]	0.074 [1.508]

(continued)

4.4 International antitrust laws, top-executive turnover, and investor protection

Bris, Cabolis, and Janowski (2010) find evidence that countries adopting antitrust laws for the first time experience an increase in aggregate M&A activity. They argue that such laws promote M&A transactions by creating a level playing field and reducing information asymmetries. We therefore use the adoption of such laws as an alternative proxy for an increase in the threat of takeover. Using the data of Bris and others (2010), we find that nine countries passed an antitrust law during our sample period. Similar to our analysis of takeover laws, we restrict the sample to countries that either passed an antitrust

Table 4
Continued

Panel B, antitrust laws, alternative clustering of standard errors, and an alternative sample

Variables	Adoption of antitrust laws		Clustering by country		12 countries only	
	Civil law	Common law	Civil law	Common law	Civil law	Common law
	(1)	(2)	(3)	(4)	(5)	(6)
Post*treat	-0.136 [-0.914]	-0.193 [-0.601]	0.384** [2.104]	0.073 [0.334]	0.271** [2.070]	-0.145 [-0.661]
Post*treat*firm performance	-0.636** [-2.125]	-0.901 [-0.403]	-1.775** [-2.450]	0.640 [1.127]	-1.821*** [-4.11]	0.562 [1.197]
Treat*firm performance	0.833** [2.099]	1.266 [0.570]	0.917 [1.249]	0.844* [1.955]	-	-
Firm performance	-0.721* [-1.870]	-0.614*** [-2.979]	-0.258 [-0.474]	-1.033** [-2.394]	0.637* [1.748]	-0.151 [-0.570]
Log assets	0.048*** [3.418]	0.040** [2.350]	0.064*** [8.662]	0.085** [2.324]	0.090*** [6.315]	0.114*** [5.093]
Inside-ownership %	0.423*** [3.289]	0.118 [0.825]	0.482*** [3.454]	0.568** [1.961]	0.456*** [3.864]	0.591*** [2.728]
Cross-listed	-0.08 [-0.652]	0.089 [0.678]	0.089 [1.531]	0.019 [0.232]	0.108 [0.757]	-0.009 [-0.045]
Log GDP per capita	0.661 [0.803]	-1.524 [-0.579]	0.099 [0.241]	0.74 [0.915]	-1.530 [-1.531]	0.851 [0.841]
Constant	-7.51 [-1.025]	13.175 [0.495]	-2.459 [-0.684]	-5.098 [-1.057]	12.627 [1.223]	-5.771 [-0.944]
Country, industry, and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	7,615	7,392	36,928	4,780	6,755	4,022
Pseudo R-squared	0.031	0.032	0.038	0.052	0.049	0.059
Mean interaction effect	-0.068** [-1.974]	-0.112 [-0.402]	-0.269*** [-4.039]	0.075 [1.360]	-0.283*** [-4.206]	0.066 [1.268]

Table 4 presents probit estimates of the effect of takeover laws on the relationship between the probability of CEO turnover and firm performance under various regression specifications. Panel A reports results from regressions where the change in foreign-institutional-ownership % is controlled for, and where alternative measures of firm performance are used, respectively. Panel B reports results from regressions where antitrust laws are used as an alternative country-level measure of the threat of takeover, the standard errors are estimated using Rogers's method of clustering by country, and for the subsample of 12 countries that adopt takeover laws for the first time during our sample period, respectively. In the first two columns in Panel B, the post- and treat-dummy definitions are based on the adoption of antitrust laws obtained from Bris and others (2010). "Change in foreign-institutional-ownership %" is the annual change in the ownership stake of institutional investors located outside the home country of the firm. "Excess stock returns" is defined as the 1-year lagged value of total investment returns in excess of the local market index, and "relative industry performance" is calculated as firm performance minus the median value of the corresponding two-digit SIC global industry. Other variables are defined in Table 1 and Table 2. Country, industry (at the 2-digit SIC level), and year fixed effects are included in every column. The z-statistics appear in brackets, below parameter estimates. Robust standard errors are estimated using Rogers's method of clustering by firm, except that Columns (3) and (4) of Panel B use clustering by country. Asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

law during the sample period (nine countries), or never passed such a law as of 2003 (fifteen countries).

To test if the enactment of antitrust laws increases the propensity to terminate poorly performing top executives, we rerun our regression specification in Equation (1) after replacing takeover laws with antitrust laws to define the post- and treat-dummy variables. Results from this estimation are reported in the first two columns of Panel B, in Table 4, for countries with civil and common law,

respectively. They show that the coefficient on the corrected triple-interaction term is negative and statistically significant in countries with weak investor protection.¹⁷

4.5 Clustering by country and the 12-country subsample

Following Petersen (2009), our results so far account for time-series and cross-sectional dependence using firm-clustered standard errors and year dummies. As an alternative method, we cluster instead by country in Panel B of Table 4 and find that our results are robust. We also replicate our tests by limiting our analysis to 12 countries that adopted takeover laws during the sample period, where the control group is firms from the same countries in the pre-law period. The final two columns in Panel B of Table 4 show that the main results are robust to this alternative sample-selection procedure. In the Online Appendix in Table 5A, we further provide 12-country summary statistics and replicate the main analyses for this subsample.¹⁸

5. Threat of Takeover, Board of Directors' Incentives and the Dismissal of Poorly Performing CEOs

Our findings show that the sensitivity of CEO turnover to poor firm-performance increases following the enactment of M&A laws around the world. As previously discussed, theory suggests a channel that can explain these findings: the threat of takeover provides additional incentives for the board of directors to fire poorly performing CEOs. To test for this channel, we first examine how the passage of M&A laws affects the threat of takeovers. Although our previous findings show that aggregate M&A activity increases following takeover-law adoption, it is not clear if this increase comes from either the acquisition of poorly performing firms (a disciplining effect) or the acquisition of profitable firms, perhaps by new foreign investors (a cherry-picking effect). If M&A laws cause a disciplining effect, we should observe that poorly performing firms are more likely to be targeted after the initiation of M&A laws. This test provides evidence on whether the probability of a takeover conditional on poor performance is greater after the passage of M&A laws. However, for the increased threat of a takeover to affect the board's diligence, there must be a penalty imposed on directors of these firms in the event of a control contest (e.g., Harford 2003). Therefore, we also examine the labor-market-related costs imposed on the directors of targeted firms where

¹⁷ We also examine the joint effect of takeover laws and antitrust laws, as it is possible that the law that was adopted earlier has the greatest influence on the turnover to performance sensitivity. When we reestimate our models after replacing the takeover laws dummy with this new variable, we continue to find that the enactment of M&A laws increases the sensitivity of CEO turnover to poor firm performance.

¹⁸ We further replicate our main regressions after excluding each one of these 12 countries one by one and we report the results in Table 6A in the Online Appendix. It shows that our results are robust to this additional test.

we test whether directors are more likely to lose their seat on the board of the targeted firm and if they experience a reduction in the number of board seats held in other firms.

5.1 International M&A laws and takeover activity

We estimate a probit regression model in which we regress the probability of being acquired on firm performance, along with a set of M&A determinants from Ferreira and others (2010). As in our previous analysis with aggregate M&A activity, we run this regression specification separately for any type of acquisitions and cross-border acquisitions. Our source of M&A transactions is the Security Data Corporation.¹⁹

Results from this estimation are reported in Table 5. The first three columns report results for any type of acquisition in the home country following the adoption of M&A laws, for the full sample, as well for the subsamples based on the English legal tradition. Column (1) shows a negative and statistically significant coefficient on the corrected triple-interaction term *post*treat*firm performance* (-0.120 , $t = -3.02$). Compared with the period prior to the adoption of takeover laws and firms in countries without such a law, the absolute probability of being targeted increases by 1.23% when we move from the top quartile to the bottom quartile of firm performance (where the average probability of being targeted across all countries is 6.48%). The second and third columns show that the probability of being acquired conditional on poor performance increases following the passage of M&A laws in civil-law countries. In Columns (4) through (6), we model the probability of a local firm being acquired by a foreign investor. Results show that some of the disciplining effect of takeover laws on poorly performing firms also arises from cross-border acquisitions, as shown by the statistically significant corrected coefficient on the triple-interaction term (-0.032 , $t = -1.71$). In robustness tests (untabulated), we repeat the analysis in Table 5 after excluding majority acquisitions, and find similar effects of M&A laws on the relation between the probability of being targeted and firm performance.

The results from Table 5 show that the threat of takeover significantly increases after the passage of M&A laws, and this threat focuses on poorly performing firms. When we split the full sample into two groups, based on the strength of investor protection, we show that the effect of the laws on M&A activity is generally limited to civil-law countries, consistent with the main results in Table 2. Overall, the results suggest the passage of M&A laws results in an increased probability that poorly performing firms will face a disciplining control contest.

¹⁹ We exclude LBOs, recapitalizations, privatizations, self-tender offers, exchange offers, spin-offs, repurchases, partial equity-stake purchases, and acquisitions where the acquirer is a government agency.

Table 5
M&A laws, firm performance, and takeover activity

Variables	All acquisitions			Cross-border acquisitions		
	Full sample (1)	Civil law (2)	Common law (3)	Full sample (4)	Civil law (5)	Common law (6)
Post*treat	0.031 [0.441]	0.004 [0.053]	-0.265** [-2.238]	0.022 [0.259]	0.026 [0.287]	-0.399** [-2.207]
Post*treat*firm performance	-1.222*** [-6.212]	-1.014*** [-7.010]	-0.795 [-0.550]	-0.664* [-1.781]	-0.540* [-1.699]	1.245 [0.866]
Treat*firm performance	1.623*** [3.031]	1.441** [2.294]	2.305 [1.594]	1.232*** [3.386]	1.286*** [4.155]	0.103 [0.072]
Firm performance	-1.263** [-2.523]	-1.126* [-1.923]	-2.654*** [-7.708]	-1.112*** [-4.854]	-0.972*** [-4.321]	-2.425*** [-4.511]
Exchange rate change (%)	0.049 [0.855]	0.016 [0.488]	1.099*** [4.093]	0.081 [1.211]	0.04 [1.173]	1.157*** [2.845]
Market return (%)	0.058* [1.756]	0.067** [1.979]	-0.0003 [-0.002]	0.066 [1.502]	0.024 [0.458]	0.144 [1.053]
Log assets	0.023** [2.244]	0.018* [1.835]	0.063*** [2.670]	0.014 [1.027]	0.009 [0.635]	0.034 [1.406]
R&D expenses/Sales	0.792 [1.071]	0.651 [0.900]	2.821 [0.784]	1.998** [1.992]	1.894** [2.033]	2.011 [0.407]
Foreign sales/sales	0.142* [1.763]	0.132 [1.426]	0.172* [1.784]	0.269*** [3.428]	0.232*** [2.784]	0.285 [1.132]
Leverage	0.117 [1.159]	0.187 [1.564]	-0.084 [-0.716]	-0.278* [-1.721]	-0.124 [-0.634]	-0.706*** [-2.834]
Inside-ownership %	0.628*** [6.128]	0.663*** [6.422]	0.434** [2.111]	0.506*** [4.267]	0.504*** [3.848]	0.498* [1.734]
Cross-listed dummy	0.094 [0.618]	0.121 [0.677]	0.094 [0.421]	0.147 [0.890]	0.290** [2.050]	-0.483 [-1.169]
Log global industry Q	-0.064* [-1.934]	-0.074** [-2.111]	0.054 [0.209]	-0.215** [-2.264]	-0.326*** [-3.017]	0.167 [0.704]
Log GDP per capita	0.114 [0.199]	0.784 [1.228]	-0.900*** [-4.578]	-0.163 [-0.231]	0.761 [1.081]	-1.345*** [-2.727]
Domestic credit/GDP	-0.046 [-0.560]	-0.114** [-2.391]	0.511* [1.878]	0.054 [0.541]	0.059 [1.617]	0.294 [0.701]
Constant	1.824 [0.321]	-4.453 [-0.678]	3.662** [2.406]	9.018 [1.345]	3.562 [0.471]	5.845* [1.769]
Country, industry, and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	38,113	33,753	4,229	38,113	33,441	3,673
Pseudo R-squared	0.132	0.128	0.133	0.158	0.159	0.145
Mean interaction effect	-0.120*** [-3.021]	-0.087** [-2.233]	-0.092 [-0.364]	-0.032* [-1.717]	-0.029 [-1.395]	0.183 [0.958]

Table 5 presents probit estimates of the effect of takeover-law adoption for the first time on the relation between the likelihood of being acquired and firm performance, separately for all M&As and cross-border M&As. The dependent variable is the likelihood of being acquired in Columns (1) through (3), and the likelihood of being acquired by a foreign acquirer in the last three columns. “Exchange rate change (%)” is the percent change in the local currency vis-à-vis \$U.S. on an annual basis. “Market return (%)” is the change in local stock market index. “R&D expenses/sales” is the ratio of R&D expenses to net sales. “Foreign sales/sales” is the ratio of foreign sales to net sales. “Log global industry Q” is the natural logarithm of the median Tobin’s Q for each cluster of firms within the same two-digit SIC code in a given year. “Domestic credit/GDP” is the ratio of domestic credit available to private industry to GDP. Other variables are defined in Table 1 and Table 2. Country, industry (at the two-digit SIC level) and year fixed effects are included in every column. The continuous variables are winsorized at the 1%- level, and all time-varying independent variables are lagged 1 year. The z-statistics appear in brackets, below parameter estimates. Robust standard errors are estimated using the Rogers method of clustering by firm. Asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

5.2 Board discipline around takeovers

We next examine the labor-market-related costs imposed on directors of firms that are targets of control contests. We conduct our tests using a sample of board members around the world, obtained from BoardEx, and merged with Worldscope. We track each director's employment within each firm over time for our analysis. We create a board member-turnover indicator whenever the director is replaced. On average, we observe a board turnover of 7.2% in our sample, similar to the 10.65% and 7.6% director turnover rates for U.S. firms reported by Asthana and Balsam (2010) and Horstmeyer (2011).

5.2.1 Director turnover around takeovers. To test if the probability of replacement is higher for board members whose firms are subject to acquisitions, we use the Securities Data Company (SDC) dataset to identify firms subject to M&A deals.²⁰ There are 381 unique firms and 2,190 unique directors (8,233 director-year observations) that we are able to match between the SDC and BoardEx datasets for the 1993-2003 period.²¹ We create event dummy variables that denote the year, the year after, 2 years after, and 3 years after the M&A deal to account for the fact that some firms have staggered boards that prevent immediate director replacement around takeovers (See, e.g., Harford 2003.). In our tests, t_0 is equal to one for firms targeted in a completed M&A deal during the year of the deal, and zero otherwise. We exclude observations corresponding to these directors after the third year.

We estimate a director-firm-level probit model in which we regress the probability of director replacement on the target indicator variables along with controls for firm performance and size; director characteristics, such as age, experience, and country; and year and industry dummies. Results from these regressions are reported in Panel A of Table 6. The first column shows that directors of target firms are more likely to be replaced in the year of takeover (t_0), compared with directors from all other firms in the BoardEx dataset for our sample countries. Column (1) also shows a higher director turnover for up to 2 years following the acquisition. In terms of economic significance, being targeted increases director turnover by 7.1% on average, after controlling for firm performance and director, country, industry, and year effects, as reported in the final row in Panel A. Given that the average director turnover is 7.2% in the

²⁰ We eliminate acquisitions labeled as privatization and repurchase, nonpublic targets, incomplete deals, and full acquisitions, and focus only on the first takeover deal in our sample period of 1993 to 2003, to analyze the effects of M&A deals on directors' career prospects. Our results are even stronger when we limit the acquisitions to those where the percent of shares acquired in the target is at least 20% or to those where the acquirer owns at least 50% of the target because of the acquisition. For example, if we follow the filter used by Kini and others (1995), which focuses on tender offers where the percent of shares acquired in the target is at least 15% and the acquirer owns less than 50% of the target prior to the acquisition, we find the probability of turnover is 9.4% higher for directors in such firms, compared with other directors in the entire sample.

²¹ Because we examine the influence of takeover activity on director replacement and future employment for up to 3 years following the M&A deal, the sample period ends in 2006 (that is, 3 years after our main sample period ends in 2003).

Table 6
The labor-market consequences to directors of being taken over

Panel A, director turnover around takeover activity

Variables	All BoardEx sample countries			Matched sample		
	Full sample (1)	Civil law (2)	Common law (3)	Full sample (4)	Civil law (5)	Common law (6)
T_0	0.084** [2.013]	0.069* [1.759]	0.260* [1.958]	0.019 [0.279]	0.037 [0.467]	0.051 [0.557]
T_1	0.288*** [9.139]	0.287*** [7.593]	0.368*** [7.977]	0.220*** [3.655]	0.204*** [3.320]	0.227** [2.163]
T_2	0.105*** [3.324]	0.119*** [3.542]	0.034 [0.662]	0.165*** [3.573]	0.164*** [3.125]	0.277*** [5.611]
T_3	0.027 [0.373]	0.034 [0.448]	-0.054 [-0.548]	0.137 [1.605]	0.169* [1.940]	-0.205* [-2.061]
Firm performance	-0.310*** [-3.362]	-0.384*** [-3.839]	-0.111** [-2.239]	-0.103 [-0.839]	-0.180 [-1.086]	0.352 [1.386]
Log assets	0.082*** [7.536]	0.086*** [7.730]	0.063*** [3.136]	0.085*** [5.592]	0.086*** [5.524]	0.083 [1.357]
Chair dummy	0.109*** [2.924]	0.082** [2.272]	0.246*** [5.603]	0.193*** [2.878]	0.192** [2.543]	0.230*** [2.723]
Log age	0.518*** [4.314]	0.543*** [4.202]	0.379* [1.792]	0.679*** [4.138]	0.703*** [3.863]	0.521* [1.875]
Log directorial experience	-0.101*** [-5.875]	-0.110*** [-6.279]	-0.048*** [-5.100]	-0.191*** [-7.609]	-0.202*** [-7.967]	-0.115** [-2.109]
Log CEO experience	-0.081*** [-7.540]	-0.082*** [-6.493]	-0.070*** [-3.963]	-0.102*** [-4.162]	-0.110*** [-4.362]	-0.050** [-2.237]
Female	-0.170*** [-3.414]	-0.161*** [-2.917]	-0.188* [-1.698]	-0.009 [-0.119]	0.012 [0.150]	-0.258 [-1.005]
Number of awards	-0.003 [-0.720]	-0.007 [-1.313]	0.005 [0.754]	0.0004 [0.050]	0.002 [0.243]	-0.004 [-1.317]
Number of committee chairs	-0.419*** [-10.042]	-0.427*** [-10.521]	-0.413*** [-4.240]	-0.458*** [-6.885]	-0.422*** [-7.636]	-0.831*** [-2.827]
Constant	-6.198*** [-9.548]	-6.563*** [-8.331]	-5.237*** [-8.378]	-5.732*** [-10.004]	-5.830*** [-9.280]	-5.600*** [-3.774]
Country and industry industry, Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	80,807	63,825	16,320	16,166	13,345	2,821
Pseudo R-squared	0.095	0.087	0.105	0.087	0.204	0.227
Average marginal effects (sum of marginal effects for T_0 through T_3)	0.071***	0.079***	0.064***	0.085**	0.096**	0.032*

(continued)

entire sample, this is an economically significant increase in director turnover. In the next two columns, we separately analyze the effects of acquisitions on director turnover in civil- and common-law countries and find that director turnover is higher in both subsamples.

For robustness, we replicate the probit estimation in the first three columns with an alternative sample of control firms. Target firms may differ from other firms in several ways, such as firm performance, industry, and size. We therefore create a control sample where we match target firms to other firms based on firm performance and size as of the year prior to an acquisition, and country and

Table 6
Continued

Panel B, the number of directorships around takeover activity

Variables	All BoardEx sample countries			Matched sample		
	Full sample (1)	Civil law (2)	Common law (3)	Full sample (4)	Civil law (5)	Common law (6)
T ₀	−0.031 [−0.599]	−0.086 [−1.333]	0.039 [0.118]	−0.147* [−1.948]	−0.181** [−2.576]	0.412 [0.594]
T ₁	−0.127** [−2.537]	−0.181*** [−2.868]	−0.110 [−0.309]	−0.170** [−2.122]	−0.195*** [−2.992]	0.463 [1.046]
T ₂	−0.219** [−2.432]	−0.201*** [−2.857]	−0.140 [−0.440]	−0.146** [−2.153]	−0.240*** [−4.863]	0.006 [0.015]
T ₃	−0.149*** [−2.993]	−0.176** [−2.260]	−0.561*** [−4.559]	−0.091 [−1.166]	−0.191** [−2.157]	0.301 [0.485]
Average firm performance	−0.304** [−2.253]	−0.397*** [−2.984]	−0.121 [−0.956]	−0.123* [−1.735]	−0.085 [−1.333]	−0.038 [−0.246]
Average log assets	0.166*** [4.652]	0.161*** [3.730]	0.171*** [8.872]	0.152*** [3.752]	0.152*** [4.556]	0.084 [1.169]
Average chair dummy	−0.021 [−0.487]	0.009 [0.230]	−0.166 [−1.611]	0.057 [0.558]	0.074 [0.682]	0.055 [0.163]
Log age	0.497*** [7.021]	0.537*** [7.046]	0.371*** [4.122]	0.426*** [4.270]	0.566*** [6.199]	0.248 [1.829]
Log directorial experience	1.019*** [3.928]	0.909*** [4.322]	1.565 [1.539]	1.059*** [4.059]	0.905*** [3.762]	2.077 [1.914]
Log CEO experience	−0.081 [−0.494]	−0.163 [−0.863]	0.064 [0.150]	0.061 [0.360]	−0.108 [−0.690]	−0.119 [−0.269]
Female	−0.277*** [−3.385]	−0.237** [−2.576]	−0.436*** [−3.896]	−0.287** [−2.224]	−0.292** [−2.643]	−1.300** [−3.035]
Number of awards	0.692*** [4.298]	0.877*** [6.782]	0.075** [2.657]	0.780*** [3.726]	1.074*** [4.625]	0.060 [0.685]
Constant	−5.551*** [−4.972]	−4.912*** [−4.487]	−6.637 [−1.743]	−6.031*** [−3.289]	−5.283*** [−3.542]	−8.132 [−1.436]
Country and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30,479	26,371	4,059	6,606	6,004	567
R-squared	0.132	0.131	0.105	0.126	0.146	0.139
Average effect (sum of T ₀ through T ₃)	−0.528**	−0.646***	−0.772	−0.556**	−0.808***	1.184

Table 6 presents probit estimates of the effect of M&A events on the probability of director turnover in target firms in Panel A and the ordinary least squares (OLS) estimates of the effect of M&A events on the number of other seats held by directors of target firms in Panel B. “Director turnover” equals one if the director is replaced by the firm in a given year, and zero otherwise. The number of directorships is the number of board seats held by target firms’ directors in other firms. Information on directorial employment history within firms and across countries on an annual basis is obtained from BoardEx. In each panel, the first three columns report results from regressions that use all the directors of firms located in our sample countries covered by BoardEx. The last three columns report results from the matched sample, where the nearest-neighbor matching procedure with the Mahalanobis metric as the weighting criterion is used to create the matched sample. The criteria used to match target firms to other firms include firm performance and size as of the year prior to an acquisition, as well as country and industry membership at the 1-digit SIC level. T₀ equals one for the year of the control event, and zero otherwise. Similarly, T₁, T₂, and T₃ equal one for 1 year, 2 years, and 3 years following the control event, and zero otherwise, respectively. Chair dummy equals one if the director is the chair of the board of directors of the firm, and zero otherwise. “Log age” is the natural logarithm of director’s age. “Log directorial experience” is the natural logarithm of director’s experience as a board member of a publicly traded firm. “Log CEO experience” is the natural logarithm of director’s experience as a top manager of a publicly traded firm. “Female” equals one if the director is female, and zero otherwise. “Number of awards” is the number of managerial awards that the director received in a given year. “Number of committee chairs” is the number of committees that the director is the chair of in the firm in a given year. Panel B uses the average values of firm performance, size, and the chair dummy across all of the firms by which the director is employed. Other variables are defined in Table 1 and Table 2. The z-statistics appear in brackets, below parameter estimates. Robust standard errors are estimated using Rogers’s method of clustering, by firm. Asterisks ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively.

industry membership (at the one-digit SIC level).²² We use the nearest-neighbor matching procedure with the Mahalanobis metric as the weighting criterion (Abadie et al. 2004). The regression sample includes 8,101 director-years from firms subject to M&A deals, and 8,065 director-years from control firms. Results from this estimation are reported in Columns (4)–(6) of Panel A. They show that directors subject to takeovers are significantly more likely to be replaced, compared with the matched sample of firms in the t -to- $t+3$ period. Even though director turnover is higher in both subsamples, the economic significance in civil-law countries is greater (9.6% versus 3.2%). Overall, this panel shows that dismissal rates are both statistically and economically higher for directors of firms targeted in M&A deals, compared with other firms' directors.²³

5.2.2 Director turnover in other firms around takeovers. We next analyze the number of board seats held by target firms' directors in other firms around M&A deals. We follow an empirical setup similar to the previous analysis, where we measure the effects of being targeted on the number of board seats in publicly traded firms within 3 years following the acquisition event. We exclude the seat lost because of the takeover within 3 years following the acquisition in these tests to measure the effect of the M&A event on the other seats held by the director. Another difference in the specifications from Panel A is that we use the average values of firm performance, size, and the chair dummy across all of the firms by which the director is employed. This is because our variable of interest is the number of board seats held by each director in a given year. Consequently, the sample size is smaller in this panel because the unit of analysis switches from director-firm-year to director-year level, as the average number of publicly traded firms that each director is associated with is 2.56 in our sample.²⁴ The first three columns focus on the entire BoardEx sample of our countries, and the last three columns use the matched sample of control firms as described earlier.

Column (1) of Panel B shows that the coefficient estimates on takeover event years 1, 2, and 3 following the M&A event are statistically significant at the 5% or lower level. Directors on average lose 0.528 seats within 3 years following

²² The implicit assumption in our matching strategy is that directors of the target have similar characteristics to those of the control firm. We opt for controlling for director characteristics such as experience, age, and education in the regression specification rather than as additional matching criteria so as not to reduce the sample size. Including the average past performance of directors across all firms they are working for and the number of board seats they hold as additional matching criteria does not change the conclusion from these regressions.

²³ We also undertake an alternative, firm-level regression specification, where we examine the effect of M&A activity on the board-turnover ratio, defined as the ratio of board members replaced in a given year to board size. In this sample, the average board-replacement ratio for the entire sample and the matched sample is 5.46% and 6.39%, respectively. We again find that the average board-replacement ratio is significantly higher in the years 0 through 2 following the acquisition for the entire sample, and years 1 through 3 for the matched sample. This effect is also more pronounced in civil law countries.

²⁴ This is consistent with the average number of directorships reported in other studies—3.11 for largest 500 U.S. firms (Fich and Shivdasani 2006); 1.94 in addition to the seat directors hold in target firms for Fortune 1000 firms (Harford 2003); and 2.9 in Field, Lowry, and Mkrtchyan (2013).

an M&A event, calculated as the sum of coefficient estimates on the event-year dummies. This number translates into an approximately 20% decrease for an average director. In the next two columns, we find that the adverse effects of being subject to a takeover on the future employment opportunities of directors are primarily observed in civil-law countries. The last three columns show results for the matched sample approach. We find that directors lose 0.556 seats on average in the 3-year period following the M&A event. This effect is even higher in civil-law countries, with an average seat loss of 0.808, whereas the event-year dummies are statistically significant only for firms in civil-law countries. Overall, Panel B shows that directors of target firms are subject to fewer opportunities for future employment as board members, highlighting another adverse consequence in directorial labor markets of being targeted.

In untabulated tests, we also run regressions where we examine the effects of pretakeover firm performance on the probability of director turnover and the number of seats held by directors for the sample of firms subject to takeovers in the t_0 -to- t_3 period. Results show that pretakeover firm performance is strongly significantly related to the dismissal probability and the number of seats held (e.g., the coefficient estimate on the t_0 *pretakeover firm performance is -1.378 , $p=0.019$).²⁵ In addition, we obtain similar results when we repeat the analysis in Table 6 after excluding majority acquisitions.

Taken together, results in Table 6 provide empirical support that directors experience adverse labor-market consequences when their firms are targeted. Board members are more likely to lose their position in the target firm, as well as their positions in other firms not associated with the target firm.²⁶ These results are consistent with theoretical predictions that the threat of takeover provides additional incentives for the board of directors to fire poorly performing CEOs because they themselves are more likely to be disciplined when their firm is taken over for poor performance.

6. Conclusions

Even though the market for corporate control is an important corporate-governance device, evidence on its effectiveness as a managerial disciplinary mechanism is mixed. This is in part because the level of takeover activity is

²⁵ Further, we run tests where we examine the effect of M&A events on director turnover and the number of directorial seats for boards that have removed CEOs in the recent past. The results, (reported in Table 7A in the Online Appendix) show that director turnover and the number of seats lost are significantly lower for such directors in two of four event-year dummies in the entire sample, and the matched sample: except that we do not observe any such effects on the number of board seats in the matched sample. These results are roughly in line with Farrell and Whidbee (2000), who find a greater likelihood of an additional directorship for directors who remove underperforming CEOs.

²⁶ These directors may suffer from additional negative effects that are difficult to quantify, such as lower future compensation and smaller networking circles.

used to proxy for the threat of takeover, and changes in managerial turnover and takeover activity can be explained jointly by unobserved macroeconomic conditions. Similarly, tests that employ takeover defenses as a proxy for the degree of takeover threat suffer from endogeneity concerns because they are often established at the discretion of the firm.

In this article, we use a different approach and examine the role that takeover markets play in disciplining poorly performing managers by using the staggered initiation of M&A laws around the world as a source of exogenous variation in the threat of takeover. We find that the enactments of these laws lead to an increased sensitivity of CEO turnover to poor firm performance. Further, our results show that M&A laws have a larger effect on the propensity to terminate poorly performing CEOs in countries with weak investor-protection laws. These results indicate that the market for corporate control works as a disciplinary governance mechanism in presence of weak corporate governance.

We also provide evidence on the board mechanism that prior studies suggest can explain the observed improvement in the sensitivity of CEO turnover to poor firm performance around M&A laws. We not only show that poorly performing firms are more likely to be targets, compared with other firms not subject to takeover laws but also document the penalties imposed on directors in the event of a control contest. We find that directors are likely to lose their board seat and might experience a reduction in the number of board seats they hold in other firms. Our findings suggest that one channel through which M&A laws increase managerial discipline in poorly performing firms is the incentives that the market for corporate control provides to boards to monitor managers.

Appendix

Table A1
Information on M&A laws

Country	Takeover-law year	Name of the law	Source
Argentina	none	-	IBA and Mazer, P. 2010. Executive view – Argentina: minority shareholders rights in mergers. http://executiveview.com/knowledge_centre.php?id=11735 .
Australia	1975	Foreign Acquisitions and Takeovers Act	Australia also established a takeovers panel in 2000 following the CLERP Act of 1999. Our results are unchanged if we use the year of 2000 instead of 1975 as the year of the takeover law.
Austria	1998	Takeover Act	IFLR, September 1998 and the 2008 M&A yearbook, and eStandards Forum.
Belgium	1989	Royal Decree of 11/8/1989	Nenova (2006) and http://www.cbfa.be/nl/oa/oa/wg/oa/wg.asp .
Brazil	none	-	ICLG.
Canada	1966	Ontario Securities Act (1966)	ICLG. In addition, the Investment Canada Act (1975) further regulates large investments and takeovers for foreign buyers.
Chile	2000	Tender Offer Act	Nenova (2006) and IBA.
China	none	-	IBA.
Colombia	none	-	ICLG and IBA.
Czech Republic	none	-	IBA and eStandards Forum.
Denmark	none	-	ICLG and Christensen, S. and A. Westenholz. 1999. Boards of directors as strategists in an enacted world – the Danish case. <i>Journal of Management and Governance</i> 3: 261–86.
Finland	1989	Takeovers panel established by the Securities Markets Act of 1989	ICLG.
France	none	-	Simmons & Simmons. EU Takeovers Directive European Implementation Proposals (available at http://www.simmons-simmons.com).
Germany	2002	Takeover Act	Gordon (2002); Baum (2006); Odenius (2008); Mondaq Business Briefing June 2002 (The M&A transactions were governed by the voluntary takeover code before 2002.).
Greece	none	-	IBA.
Hong Kong	1975	Code on takeovers and mergers	Nenova (2006) and http://www.sfc.hk/sfcPressRelease/EN/ .
Hungary	none	-	Budai, Judit. 2007. Hungary. The public takeover bid – assessment and risks. http://www.mondaq.com/article.asp?articleid=47136 .
India	1997	Substantial Acquisition of Shares and Takeovers	IBA, ICLG, and Securities and Exchange Board of India.
Indonesia	1998	M&A Regulations (Government regulation No. 27/1998 and Presidential Decree No. 96 and No. 118)	Haecys Associates, Indonesian Acquisition Law in a nutshell (available at http://www.bakermckenzie.com), and http://www.haecys.com/uploadedfiles/library/file_01183020030415.pdf .
Ireland	1997	Takeover Panel Act	IFLR. 2007 M&A yearbook, and Given, J. and C. McCourt. 2005. Ireland. In <i>Mergers and acquisitions</i> : 2005/06, London: Practical Law Co.
Israel	none	-	Nenova (2006).

(continued)

Table A1
Continued

Italy	1992	Public Tender Offer	ICLG and Simmons & Simmons. EU Takeovers Directive European Implementation Proposals. IBA. ICLG. http://www.legilux.public.lu/leg .
Japan	none	—	
Korea	none	—	
Luxembourg	none	—	
Malaysia	1998	Code on takeovers and mergers	
Mexico	none	—	Nenova (2006) and ILFR 2002 M&A yearbook. http://www.worldbank.org/ifa/rosc_cg_mex.pdf p53. http://www.sernl/~media/Files/Internet/ .
Netherlands	1970	Merger Code of the Social Economic Council	
New Zealand	2001	Takeovers Code	http://www.bellgully.com/resources/resource/00063.asp . http://www.legislation.govt.nz/act/public/1993/0107/latest/DLM325810.html#DLM325810 . In addition, New Zealand passed a legislation on Takeovers Panel Advisory Committee in 1991.
Norway	none	—	Sjåfjell, B. 2006. Country report from Norway: The new takeover regime (2). <i>European Company Law</i> 3: 202–06; Ibanet and ICLG.
Pakistan	2000	Ordinance on substantial acquisition of shares and takeovers of listed companies	Nenova (2006), Ibrahim, A. 2006. Corporate governance in Pakistan: Analysis of current challenges and recommendations for future reforms. <i>Washington University Global Studies Law Review</i> 5: 323–32; Khan & Associates.
Peru	none	—	IBA.
Philippines	1998	Tender Offer Rules	IFLR, November 1998.
Poland	none	—	IBA.
Portugal	none	—	Simmons & Simmons, EU Takeovers Directive European Implementation Proposals (available at http://www.simmons-simmons.com).
Singapore	1974	Code on Takeovers and Mergers	Consultation paper on revisions of the Singapore Code n takeovers and mergers," Securities Industry Council, November 1999.
South Africa	1991	Code on Takeovers and Mergers	Nenova (2006) and http://www.gt.co.za/Publications/Effective-directors-guide/takeovers.asp . http://www.bde.es/infornes/be/boleco/2007oct/art6.pdf . http://www.perezllorca.com/include_mav/gefile.asp?dFileImage=1926 .
Spain	1991	Public Takeover Offerings (Royal Decree 1197/1991)	
Sri Lanka	1995	Company Takeovers and Mergers Code	Nenova (2006); IBA; Marsoof, Saleem, "Takeover offers and their ramifications," <i>Corporate Law: Issues in Focus</i> , available at http://www.lawnet.lk/docs/articles/sri_lankan/HTML/CV40.html .

(continued)

Table A1
Continued

Sweden	1991	Industry and Commerce Stock Exchange Committee Takeover Standard, Financial Instruments Trading Act	Nenova (2006).
Switzerland	2004	Federal Act on Merger, Demerger, Transformation, and Transfer of Assets (The Merger Act)	IFLR and IBA. The Swiss Takeover Board was established under the Federal Act on Stock Exchanges and Securities Trading in 1995, but it had no legal authority (using 1995 as the M&A law for Switzerland does not alter our conclusions).
Taiwan	2002	Business Mergers and Acquisitions Act	IFLR, 2005 and 2007 M&A yearbook, Legal Media Group's 2005 edition of Taiwan.
Thailand	none	—	Nenova (2006) and eStandards Forum.
Turkey	none	—	ICGL, and Turkey, takeover guide. http://www.thetakeoverpanel.org.uk .
United Kingdom	1968	City Code on Takeovers and Mergers	Malatesta, P., and R. Thompson. 1993. Government regulation and structural change in the corporate acquisitions market. The impact of the Williams Act. <i>Journal of Financial and Quantitative Analysis</i> 28: 363–79.
United States	1968	Williams Act	eStandards Forum. http://www.worldbank.org/ifa/rosc_cg_zimbabwe.html .
Venezuela	None	—	
Zimbabwe	None	—	

Table A1 presents detailed information on takeover laws as of the end of 2003 for each country in our sample. When available, we use existing studies to identify the presence and initial enactment year for takeover laws (e.g., Nenova 2006). For the remaining countries, we use the national regulatory websites, the takeover guides published by the International Bar Association (IBA), the International Comparative Legal Guide Series (ICLG) on Mergers & Acquisitions, International Financial Law Review (IFLR), eStandards Forum, and Factiva searches to determine the takeover-law status in each country.

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